# Core Link Technologies – Full Documentation

## 1. Executive Summary

This document provides a complete technical design and configuration reference for Core Link Technologies network including a Headquarters (HQ) and Branch (BR1), interconnected through ISP routers using public IPs, static routing, RIP, and a GRE tunnel.

The design includes a full three-tier switching architecture (Core, Aggregation, Access), VRRP high availability, LACP EtherChannel redundancy, wireless infrastructure with ACs and AP groups, DHCP services, AAA security, and IP addressing schemas for all VLANs.

## 2. Network Overview

The network includes two main sites: Headquarters (HQ) and Branch (BR1).

Connectivity between sites is achieved via ISP routers and a GRE tunnel carried over public IP addressing.

HQ uses dual core switches with full VRRP gateway redundancy and LACP-based inter-switch uplinks.

Each site features VLAN segmentation for HR, IT, CCTV, VOIP, Guest, Servers, Management, IoT, and Backup.

Wireless services are centrally managed through dual ACs (AC-HQ and AC-Suez) and CAPWAP AP deployment.

## 3. Physical & Interface Mapping

A mapping of all inter-device connections including router-to-router, core-to-aggregation, aggregation-to-access, and access-to-AP infrastructure.

All EtherChannel (Eth-Trunk) bundles are documented, including port assignments and uplink distribution.

# 🏢 HQ Site

## HQ Core

| Device | Interface | Connected To | Interface | Notes |
|---|---|---|---|---|
| HQ-R1 | G0/0/0 | HQ-1 | G0/0/1 | |
| HQ-R1 | G0/0/1 | ISP1 | G0/0/0 | |
| HQ-1 | G0/0/24 | Core-SW1 | G0/0/22 | |
| HQ-1 | G0/0/23 | Core-SW2 | G0/0/22 | |
| Core-SW1 | G0/0/22 | HQ-1 | G0/0/24 | |
| Core-SW1 | G0/0/23,24 | Core-SW2 | G0/0/23,24 | EtherChannel |
| Core-SW1 | G0/0/19,20 | Agg-SW1 | G0/0/23,24 | EtherChannel |
| Core-SW1 | G0/0/17,18 | Agg-SW2 | G0/0/23,24 | EtherChannel |
| Core-SW1 | G0/0/15,16 | Agg-SW3 | G0/0/23,24 | EtherChannel |
| Core-SW2 | G0/0/23,24 | Core-SW1 | G0/0/23,24 | EtherChannel |
| Core-SW2 | G0/0/19,20 | Agg-SW3 | G0/0/21,22 | EtherChannel |
| Core-SW2 | G0/0/17,18 | Agg-SW2 | G0/0/21,22 | EtherChannel |
| Core-SW2 | G0/0/15,16 | Agg-SW1 | G0/0/21,22 | EtherChannel |

## HQ Aggregation & Access

| Device | Interface | Connected To | Interface | Notes |
|---|---|---|---|---|
| Agg-SW1 | G0/0/23,24 | Core-SW1 | G0/0/19,20 | EtherChannel |
| Agg-SW1 | G0/0/21,22 | Core-SW2 | G0/0/15,16 | EtherChannel |
| Agg-SW2 | G0/0/23,24 | Core-SW1 | G0/0/17,18 | EtherChannel |
| Agg-SW2 | G0/0/21,22 | Core-SW2 | G0/0/17,18 | EtherChannel |
| Agg-SW3 | G0/0/23,24 | Core-SW1 | G0/0/15,16 | EtherChannel |
| Agg-SW3 | G0/0/21,22 | Core-SW2 | G0/0/19,20 | EtherChannel |
| Agg-SW3 | G0/0/19,20 | AC1 | G0/0/7,8 | EtherChannel |
| Agg-SW3 | G0/0/3 | VoIP Server | — | |
| Agg-SW3 | G0/0/2 | NVR Server | — | |
| Agg-SW3 | G0/0/1 | File Storage | — | |
| AC1 | G0/0/7,8 | Agg-SW3 | G0/0/19,20 | EtherChannel |
| AC1 | G0/0/2 | AP3 | G0/0/4 | |
| AC1 | G0/0/4 | AP1 | G0/0/4 | |
| AC1 | G0/0/6 | AP2 | G0/0/4 | |

# Branch Site (BR1)

| Device | Interface | Connected To | Interface | Notes |
|---|---|---|---|---|
| BR1-R1 | G0/0/1 | ISP3 | G0/0/1 | |
| BR1-R1 | G0/0/0 | BR-1 | G0/0/1 | |
| BR-1 | G0/0/23 | BR1-CoreSwitch1 | G0/0/16 | |
| BR-1 | G0/0/24 | BR1-CoreSwitch2 | G0/0/16 | |
| CoreSwitch1 | G0/0/21,22 | CoreSwitch2 | G0/0/21,22 | EtherChannel |
| CoreSwitch1 | G0/0/17,18 | AG-SW-1 | G0/0/23,24 | EtherChannel |
| CoreSwitch1 | G0/0/19,20 | AG-SW-2 | G0/0/21,22 | EtherChannel |
| CoreSwitch2 | G0/0/21,22 | CoreSwitch1 | G0/0/21,22 | EtherChannel |
| CoreSwitch2 | G0/0/17,18 | AG-SW-1 | G0/0/21,22 | EtherChannel |
| CoreSwitch2 | G0/0/19,20 | AG-SW-2 | G0/0/21,22 | EtherChannel |
| AG-SW-1 | G0/0/23,24 | CoreSwitch1 | G0/0/17,18 | EtherChannel |
| AG-SW-1 | G0/0/21,22 | CoreSwitch2 | G0/0/17,18 | EtherChannel |
| AG-SW-2 | G0/0/23,24 | CoreSwitch2 | G0/0/19,20 | EtherChannel |
| AG-SW-2 | G0/0/21,22 | CoreSwitch1 | G0/0/20,19 | EtherChannel |
| AG-SW-2 | G0/0/16 | AC2 | G0/0/8 | |
| AG-SW-2 | G0/0/17 | File Storage 2 | — | |
| AG-SW-2 | G0/0/18 | NVR Server 2 | — | |
| AG-SW-2 | G0/0/19 | VoIP Server 2 | — | |
| AC2 | G0/0/4 | AP7 | G0/0/0 | |
| AC2 | G0/0/6 | AP6 | G0/0/0 | |

### ISP Chain

| Device | Interface | Connected To | Interface |
|---|---|---|---|
| ISP1 | G0/0/0 | HQ-R1 | G0/0/1 |
| ISP1 | G0/0/1 | ISP2 | G0/0/0 |
| ISP2 | G0/0/0 | ISP1 | G0/0/1 |
| ISP2 | G0/0/1 | ISP3 | G0/0/0 |
| ISP3 | G0/0/0 | ISP2 | G0/0/1 |
| ISP3 | G0/0/1 | BR1-R1 | G0/0/1 |

## 4. VLAN and IP Addressing Plan

VLANs include Management, IT, HR, CCTV, Guest, VOIP, Core Applications, Servers, DMZ, IoT, Backup, and WLAN SSIDs.

Each VLAN is assigned a subnet and VRRP virtual gateway on the core switches for redundancy.

DHCP pools are configured on Core-SW1 and Core-SW2 (HQ) and similarly on BR1 cores.
Here's your HQ VLAN configuration arranged in a clean, structured table:

## 🏢 HQ VLAN Configuration Table

| VLAN ID | Subnet | Default Gateway | Purpose |
|---|---|---|---|
| 10 | 10.0.0.0/24 | 10.0.0.1 | Network device management (switches, routers) |
| 20 | 10.0.1.0/24 | 10.0.1.1 | Network infra (NTP, DNS internal, AD DS replicas) |
| 30 | 10.0.2.0/23 | 10.0.2.1 | Application servers, DB, internal services (HQ primary) |
| 40 | 10.0.4.0/24 | 10.0.4.1 | Public-facing services (webmail, reverse-proxy, APIs) |
| 50 | 10.0.5.0/24 | 10.0.5.1 | Core business applications (ERP, finance) |
| 60 | 10.0.6.0/24 | 10.0.6.1 | HR systems / restricted systems |
| 70 | 10.0.7.0/24 | 10.0.7.1 | IT engineers / admin workstations |
| 80 | 10.0.8.0/24 | 10.0.8.1 | IP Phones (native voice VLAN) |
| 90 | 10.0.9.0/24 | 10.0.9.1 | IP cameras / video storage |
| 100 | 10.0.10.0/24 | 10.0.10.1 | Guest WiFi / internet-only clients |
| 110 | 10.0.11.0/24 | 10.0.11.1 | Dev/QA lab / sensitive environments |
| 120 | 10.0.12.0/24 | 10.0.12.1 | AP management and controller traffic |
| 200 | 10.0.20.0/24 | 10.0.20.1 | Printers, IoT sensors, building automation |
| 250 | 10.0.21.0/24 | 10.0.21.1 | Backup network / replication |

Here’s the VLAN configuration for Branch 1 arranged in a structured table:

### Branch 1 VLAN Configuration Table

| VLAN ID | Subnet | Default Gateway | Purpose |
| --- | --- | --- | --- |
| 10 | 10.2.0.0/24 | 10.2.0.1 | Network device management (switches, routers) |
| 20 | 10.2.1.0/24 | 10.2.1.1 | Network infra (NTP, DNS internal, AD DS replicas) |
| 30 | 10.2.2.0/23 | 10.2.2.1 | Application servers, DB, internal services (branch servers) |
| 40 | 10.2.4.0/24 | 10.2.4.1 | Public-facing services (branch proxy, local apps) |
| 50 | 10.2.5.0/24 | 10.2.5.1 | Core business applications (ERP access, finance) |
| 60 | 10.2.6.0/24 | 10.2.6.1 | HR systems / restricted systems |
| 70 | 10.2.7.0/24 | 10.2.7.1 | IT engineers / admin workstations |
| 80 | 10.2.8.0/24 | 10.2.8.1 | IP Phones (native voice VLAN) |
| 90 | 10.2.9.0/24 | 10.2.9.1 | IP cameras / video storage |
| 100 | 10.2.10.0/24 | 10.2.10.1 | Guest WiFi / internet-only clients |
| 110 | 10.2.11.0/24 | 10.2.11.1 | Dev/QA lab / sensitive environments |
| 120 | 10.2.12.0/24 | 10.2.12.1 | AP management and controller traffic |
| 200 | 10.2.20.0/24 | 10.2.20.1 | Printers, IoT sensors, building automation |
| 250 | 10.2.21.0/24 | 10.2.21.1 | Backup network / replication |

## 5. Detailed Configuration Documentation

• VLAN creation and interface VLANIF addressing

• VRRP redundancy configuration for all gateway VLANs Each VLAN interface includes VRRP setup:

vrrp vrid <VLAN_ID> virtual-ip <Gateway IP>

vrrp vrid <VLAN_ID> priority 120

Example for VLAN 10:

```
interface Vlanif10
 ip address 10.2.0.2 255.255.255.0
 vrrp vrid 10 virtual-ip 10.2.0.1
 vrrp vrid 10 priority 120
```

• LACP-based Eth-Trunk uplinks between Core–Agg–Access layers

• LLDP for discovery and topology visibility

Enabled globally on all switches:

```
lldp enable
```

This protocol support topology visibility and neighbor discovery.

• DHCP pool creation and relay where required

```
ip pool vlanXX-sw1
 gateway-list 10.2.X.1
 network 10.2.X.0 mask 255.255.255.0
 excluded-ip-address 10.2.X.127 10.2.X.254
 dns-list 8.8.8.8 1.1.1.1
```

Relay enabled on VLAN 70:

```
interface Vlanif70
 dhcp select relay
```

• GRE tunnel configuration on BR1-R1 router

```
interface Tunnel X
 tunnel-protocol gre
 ip address <Tunnel_IP> <Mask>
```

GRE is used for secure inter-site routing (e.g., HQ ↔ BR1).

• RIP routing with version 2 and static route redistribution

```
rip 1
version 2
network <subnet>
```

Static route:

```
ip route-static X.X.X.X x.x.x.x <next-hop>
```

• AAA authentication and SSH security hardening

authentication-scheme default

authentication-scheme radius

authentication-mode radius

authorization-scheme default

accounting-scheme default

domain default

authentication-scheme radius

radius-server default

domain default_admin

authentication-scheme default

local-user admin password irreversible-cipher depi@r4

local-user admin service-type http

SSH Security

stelnet ssh server enable

ssh server secure-algorithms cipher aes256_ctr aes128_ctr

ssh server key-exchange dh_group14_sha1

• WLAN configuration including SSID, VAP, security profiles, and AP provisioning

SSID & Security

ssid-profile Corelink-SSID ssid "Corelink Technologies" security-profile Corelink-Sec security wpa-wpa2 psk pass-phrase

VAP & AP Group

vap-profile Corelink-VAP

service-vlan vlan-id 100

ssid-profile Corelink-SSID

security-profile Corelink-Sec

ap-group CORELINK-APs

radio 0

vap-profile Corelink-VAP wlan 1

radio 1

vap-profile Corelink-VAP wlan 1

AP Provisioning

ap-id 1 ap-name Corelink-AP1 ap-group CORELINK-APs

ap-id 2 ap-name Corelink-AP2 ap-group CORELINK-APs

ap-id 3 ap-name Corelink-AP3 ap-group CORELINK-APs

## 6. Routing Design

HQ and Branch are connected via a GRE tunnel established between BR1-R1 and HQ router.

Static routing ensures default route propagation to ISP next-hop.

RIP version 2 is used internally between routers to advertise internal networks through the GRE tunnel.

Routing table convergence allows full reachability between HQ subnets and BR1 subnets.

## 7. Redundancy & High Availability

Dual Core switches at HQ and BR1 use VRRP for gateway redundancy.

LACP EtherChannels provide link-level redundancy between core, aggregation, and access layers.

RSTP ensures loop-free topology with fast convergence.

Wireless APs are dual-registered across AC-HQ and AC-Suez for failover.

## 8. Wireless Network Design

Wireless network uses Huawei ACs for centralized management.

SSID 'Corelink Technologies' is broadcast using WPA2-PSK security.

APs are grouped under CORELINK-APs group with both radios broadcasting same SSID.

CAPWAP tunnels connect APs to AC-HQ or AC-Suez depending on site.

## 9. Security Architecture

AAA used for admin authentication on network devices.

SSH secure algorithms set (AES, SHA256).

Illegal MAC address alarms enabled across all switches.

STP edge-port disabled on trunks to prevent rogue STP influence.

DHCP exclusion ranges used to protect infrastructure addresses.

## 10. Testing & Validation Procedures

Connectivity Tests:

• Ping between HQ and Branch via GRE tunnel.

```
<HQ-R1>ping 10.2.0.10
  PING 10.2.0.10: 56  data bytes, press CTRL_C to break
    Reply from 10.2.0.10: bytes=56 Sequence=1 ttl=255 time=20 ms
    Reply from 10.2.0.10: bytes=56 Sequence=2 ttl=255 time=30 ms
    Reply from 10.2.0.10: bytes=56 Sequence=3 ttl=255 time=30 ms
    Reply from 10.2.0.10: bytes=56 Sequence=4 ttl=255 time=30 ms
    Reply from 10.2.0.10: bytes=56 Sequence=5 ttl=255 time=30 ms

  --- 10.2.0.10 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/28/30 ms
```

```
<BR1-R1>
<BR1-R1>ping 10.0.0.10
  PING 10.0.0.10: 56  data bytes, press CTRL_C to break
    Reply from 10.0.0.10: bytes=56 Sequence=1 ttl=255 time=30 ms
    Reply from 10.0.0.10: bytes=56 Sequence=2 ttl=255 time=30 ms
    Reply from 10.0.0.10: bytes=56 Sequence=3 ttl=255 time=40 ms
    Reply from 10.0.0.10: bytes=56 Sequence=4 ttl=255 time=30 ms
    Reply from 10.0.0.10: bytes=56 Sequence=5 ttl=255 time=30 ms

  --- 10.0.0.10 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 30/32/40 ms
```

```
<BR1-R1>tracert 10.0.0.10

 traceroute to  10.0.0.10(10.0.0.10), max hops: 30 ,packet length: 40,press CTRL
_C to break

 1 172.16.100.1 70 ms  20 ms  30 ms
```

• Ping between VLANs inside HQ and BR1.

Pc 1 from HQ to FTP server in BR1

```
PC>ping 10.2.1.252

Ping 10.2.1.252: 32 data bytes, Press Ctrl_C to break
From 10.2.1.252: bytes=32 seq=1 ttl=251 time=187 ms
From 10.2.1.252: bytes=32 seq=2 ttl=251 time=157 ms
From 10.2.1.252: bytes=32 seq=3 ttl=251 time=172 ms
From 10.2.1.252: bytes=32 seq=4 ttl=251 time=141 ms
From 10.2.1.252: bytes=32 seq=5 ttl=251 time=172 ms

--- 10.2.1.252 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 141/165/187 ms
```

• DHCP address assignment validation in all VLANs.

PC1

Basic Config | Command | MCPacket | UdpPacket | Console

```
PC>ipconfig

Link local IPv6 address...........: fe80::5689:98ff:fe51:42ff
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 10.0.7.126
Subnet mask.......................: 255.255.255.0
Gateway...........................: 10.0.7.1
Physical address..................: 54-89-98-51-42-FF
DNS server........................: 8.8.8.8
                                    1.1.1.1

PC>ipconfig

Link local IPv6 address...........: fe80::5689:98ff:fe51:42ff
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 10.0.7.126
Subnet mask.......................: 255.255.255.0
Gateway...........................: 10.0.7.1
Physical address..................: 54-89-98-51-42-FF
DNS server........................: 8.8.8.8
                                    1.1.1.1

PC>
```

Failover Tests:

• VRRP Master/Backup role switching.

```
<Core-SW1>display vrrp brief
VRID  State        Interface                Type     Virtual IP
----------------------------------------------------------------
10    Master       Vlanif10                 Normal   10.0.0.1
20    Master       Vlanif20                 Normal   10.0.1.1
30    Master       Vlanif30                 Normal   10.0.2.1
40    Master       Vlanif40                 Normal   10.0.4.1
50    Master       Vlanif50                 Normal   10.0.5.1
60    Master       Vlanif60                 Normal   10.0.6.1
70    Master       Vlanif70                 Normal   10.0.7.1
80    Master       Vlanif80                 Normal   10.0.8.1
90    Master       Vlanif90                 Normal   10.0.9.1
100   Master       Vlanif100                Normal   10.0.10.1
110   Master       Vlanif110                Normal   10.0.11.1
120   Master       Vlanif120                Normal   10.0.12.1
200   Master       Vlanif200                Normal   10.0.20.1
250   Master       Vlanif250                Normal   10.0.21.1
----------------------------------------------------------------
Total:14    Master:14    Backup:0     Non-active:0
<Core-SW1>
```

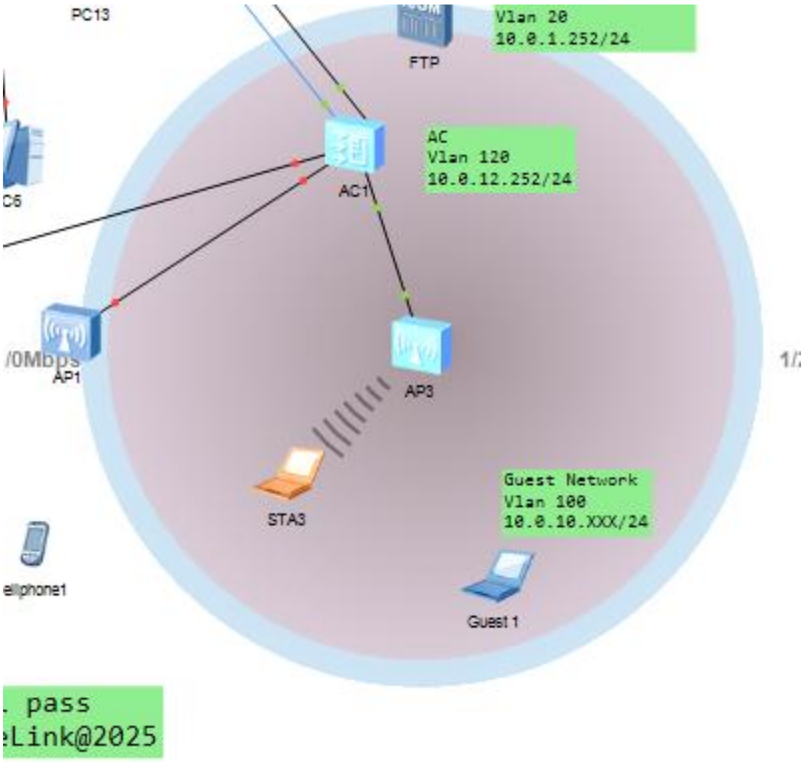
```
----------------------------------------------------------------
Total:14    Master:0     Backup:14    Non-active:0
<Core-SW2>dis vrrp br
VRID  State        Interface                Type     Virtual IP
----------------------------------------------------------------
10    Backup       Vlanif10                 Normal   10.0.0.1
20    Backup       Vlanif20                 Normal   10.0.1.1
30    Backup       Vlanif30                 Normal   10.0.2.1
40    Backup       Vlanif40                 Normal   10.0.4.1
50    Backup       Vlanif50                 Normal   10.0.5.1
60    Backup       Vlanif60                 Normal   10.0.6.1
70    Backup       Vlanif70                 Normal   10.0.7.1
80    Backup       Vlanif80                 Normal   10.0.8.1
90    Backup       Vlanif90                 Normal   10.0.9.1
100   Backup       Vlanif100                Normal   10.0.10.1
110   Backup       Vlanif110                Normal   10.0.11.1
120   Backup       Vlanif120                Normal   10.0.12.1
200   Backup       Vlanif200                Normal   10.0.20.1
250   Backup       Vlanif250                Normal   10.0.21.1
----------------------------------------------------------------
Total:14    Master:0     Backup:14    Non-active:0
<Core-SW2>
```

Wireless Tests:

• AP registration on AC.

```
<AC-HQ>dis ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor   : normal          [3]
--------------------------------------------------------------------------------
---------------------
ID   MAC            Name         Group        IP           Type            State
STA Uptime
--------------------------------------------------------------------------------
---------------------
1    00e0-fc69-4940 Corelink-AP1 CORELINK-APs 10.0.12.151 AP2050DN        nor
0   16S
2    00e0-fcd4-63e0 Corelink-AP2 CORELINK-APs 10.0.12.209 AP2050DN        nor
0   21S
3    00e0-fcba-76d0 Corelink-AP3 CORELINK-APs 10.0.12.139 AP2050DN        nor
1   4M:15S
--------------------------------------------------------------------------------
---------------------
Total: 3
```

• SSID visibility and client connectivity.



```
Welcome to use STA Simulator!

STA>ipconfig

Link local IPv6 address...........: ::
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 10.0.10.250
Subnet mask.......................: 255.255.255.0
Gateway...........................: 10.0.10.1
Physical address..................: 54-89-98-7D-19-AD
DNS server........................: 8.8.8.8
                                    1.1.1.1
```

## 11. Troubleshooting Guide

Common Issues and Fixes:

• GRE Tunnel Down: Check source/destination public IPs and route to remote endpoint.

• VRRP Not Master: Verify priority settings and interface status.

• LACP Mismatch: Ensure trunk ports have identical configuration on both ends.

• DHCP Not Assigning: Confirm interface VLANIF DHCP select global/relay.

• AP Not Joining: Check CAPWAP source IP and routing to AC.

## 12. Maintenance Procedures

• Regular backup of switch and router configurations.

• Monitor LLDP/NDP/NTDP for topology changes.

• Verify VRRP state regularly for high availability.

• Apply firmware updates during maintenance windows.